Laasya M. Koduru

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Research Interests

Reliable and cost-efficient LLM and computer-use agents for large-scale web measurement, applied to data-driven broadband policy.

Education

University of California, Santa Barbara — Santa Barbara, CA
Ph.D., Computer Science — Sept 2024–Present
M.S., Computer Science — Sept 2023–Present
Advisors: Dr. Arpit Gupta, Dr. Elizabeth Belding

University of California, Santa Cruz — Santa Cruz, CA
B.S., Applied Mathematics — Sept 2020–June 2023

Skills

- **Programming Languages:** Python, JavaScript, TypeScript, Java, C++, HTML, CSS
- **Database Systems:** PostgreSQL
- **AI, Data Science & Machine Learning:** PyTorch, LangChain, Pandas, NumPy, Matplotlib, BeautifulSoup4, Selenium, PyAutoGUI
- **Cloud Technologies:** AWS
- **DevOps & Infrastructure:** Docker, Kubernetes, Git, RabbitMQ
- **Research Skills:** AI agents for web automation, broadband and Internet measurement, large-scale data collection pipelines

Selected Publications

[1] **Laasya Koduru**, Kira Allmann, Tejas Narechania, Arpit Gupta, "From Measurement to Policy Design: A Convex Optimization Framework for Broadband Affordability", Research Conference on Communications, Information and Internet Policy (TPRC), 2026.

[2] **Laasya Koduru**, Roman Beltiukov, Alexander Nguyen, Eugene Vuong, Jaber Daneshamooz, Tejas Narechania, Elizabeth Belding, Arpit Gupta, "Robust and Extensible Measurement of Broadband Plans with BQT+", arXiv preprint, Feb 2026.

[3] Jaber Daneshamooz, Eugene Vuong, **Laasya Koduru**, Sanjay Chandrasekaran, Arpit Gupta, "NetGent: Agent-Based Automation of Network Application Workflows", Workshop on Machine Learning for Systems at NeurIPS, Dec 2025.

[4] **Laasya Koduru**, Alejandro Alvarado Rojas, Angel Chavez Penate, Siyi Zhou, François Bar, Elizabeth Belding, Hernan Galperin, Arpit Gupta, "Assessing the Broadband Service Gaps and Affordability Barriers in BEAD-Eligible Areas", Research Conference on Communications, Information and Internet Policy (TPRC), 2025.

[5] **Laasya Koduru**, Arpit Gupta, Elizabeth Belding, Tejas Narechania, "Evaluating the Effects of and Interdependencies Among Federal Broadband Funding Programs", Research Conference on Communications, Information and Internet Policy (TPRC), 2025.

[6] Zhuowei Wen, Jocelyn Bliton, **Laasya Koduru**, Arpit Gupta, Shaddi Hasan, "Strategic Misreporting in the National Broadband Map", Research Conference on Communications, Information and Internet Policy (TPRC), 2025.

[7] Haarika Manda, Varshika Srinivasavaradhan, **Laasya Koduru**, Kevin Zhang, Xuanhe Zhou, Udit Paul, Elizabeth Belding, Arpit Gupta, Tejas Narechania, "The Efficacy of the Connect America Fund in Addressing US Internet Access Inequities", Proceedings of the ACM SIGCOMM 2024 Conference.

Policy Impact

- **Congressional Brief:** Provided the BQT+ data and analysis behind a briefing to congressional staff on Capitol Hill by my advisor, a UC Presidential Faculty Fellow.
- **Virginia Joint Commission on Technology and Science (JCOTS):** BQT+ data informed the affordability benchmarks in "Beyond Access: Broadband Affordability & Adoption" (co-author), Nov 2025.
- **New York City Office of Technology and Innovation (OTI):** BQT+ address-level availability and affordability analysis informed the City's Internet Access and Adoption Plan and its fiscal-impact assessment (co-author).
- **California Public Advocates Office (CalAdvocates), CPUC:** Contributed BQT pricing data to "Broadband Competition and Pricing Strategies in California's Urban Markets" (2026), whose analysis found over $1B per year in potential consumer savings.

Honors

- UC Santa Barbara Neal Fenzi Resonant Founder Fellowship — April 2026–Present
- The National GEM Consortium Ph.D. Science and Engineering Fellowship — April 2025–Present
- IETF/IRTF Applied Networking Research Prize (SIGCOMM 2024 Connect America Fund paper) — April 2025
- UC Santa Barbara Computer Science Summer Fellowship — June 2024

Experience

University of California, Santa Barbara — Santa Barbara, CA
Graduate Student Researcher — Sept 2023–Present
Systems & Networking Lab (SNL) and Mobility Management & Networking Lab (MOMENT)

- Designed BQT+, an agentic system in which frontier LLMs discover each ISP's address-checker workflow from a natural-language goal and emit a reusable workflow specification. Covers 100+ ISPs.
- Replays cached specifications across addresses at scale, with 100% agreement on 500 address–ISP pairs. Measured 56 website changes across 64 ISPs in eight months.
- Released public broadband plan data and a natural-language query interface used by Virginia JCOTS, NYC OTI, and CalAdvocates.
- Deployed BQT+ on Kubernetes in Google Cloud with autoscaling and load balancing across servers.

Idaho National Laboratory — Idaho Falls, ID
Cybersecurity Machine Learning Algorithms Intern — June–Sept 2025

- Built a bidirectional LSTM model using PyTorch and TensorFlow that separates mixed Ethernet communication signals into their original source signals, reaching 96% accuracy against a 90% baseline.
- Tuned 200+ hyperparameter configurations on INL's HPC cluster.

Projects

CVE-Genie: Automated CVE Reproduction | *Python* — Sept–Dec 2024

- Helped build an LLM multi-agent system that rebuilds, exploits, and verifies CVEs from public reports; reproduced 50 CVEs end to end, including C/C++ memory-safety bugs.

MOVERC: MongoDB Version Classifier | *Python* — April–June 2024

- Built a random forest that infers MongoDB versions (3.0–7.0) from packet headers to flag vulnerable servers, reaching 0.74 weighted F1 vs. 0.42 for a neural net baseline on a 30K-flow dataset.